

Experiment No: 1.Introduction to Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup.

Aim: The aim of this experiment is to introduce Maven and Gradle as build automation tools, highlighting their key differences, and providing a hands-on guide to their installation and setup.

Procedure:

Step 1: Introduction to Build Automation Tools

These tools automate repetitive tasks like compiling, testing, packaging, and deploying software. Maven and Gradle are two popular tools used for this purpose in Java-based projects.

- **Maven:** An XML-based tool that follows a convention-over-configuration principle.
- **Gradle:** A more flexible build tool that uses Groovy or Kotlin for configuration and is designed for higher performance.

Step 2: Key Differences Between Maven and Gradle

- **Configuration:**
 - Maven uses XML for configuration (pom.xml).
 - Gradle uses Groovy or Kotlin-based DSL for configuration (build.gradle or build.gradle.kts).
- **Performance:**
 - Gradle tends to be faster due to incremental builds and parallel execution.
 - Maven follows a more rigid build process that can be slower.
- **Flexibility:**
 - Gradle is highly flexible, allowing custom plugins and task configurations.
 - Maven offers a structured and standardized approach but is less flexible.
- **Dependency Management:**
 - Both tools manage dependencies but Maven uses repositories and Gradle leverages a more powerful and flexible dependency resolution system.

Step 3: Installing Maven and Gradle on Ubuntu 24.04

Install Maven

1. Update package index:

```
sudo apt update
```

2. Install Maven:

```
sudo apt install maven
```

3. Verify Maven installation:

```
mvn -v
```

Install Gradle

1. Install required dependencies:

```
sudo apt install openjdk-11-jdk wget
```

2. Download Gradle:

```
wget https://services.gradle.org/distributions/gradle-7.5.1-bin.zip
```

3. Unzip the downloaded Gradle archive:

```
sudo unzip gradle-7.5.1-bin.zip -d /opt
```

4. Set up environment variables:

- Open the `.bashrc` file for editing:

```
nano ~/.bashrc
```

- Add the following lines to set the `GRADLE_HOME` environment variable and update the `PATH`:

```
export GRADLE_HOME=/opt/gradle-7.5.1
export PATH=$PATH:$GRADLE_HOME/bin
```

- Save and close the file, then reload `.bashrc`:

```
source ~/.bashrc
```

5. Verify Gradle installation:

```
gradle -v
```

Outputs:

Install Maven:

Output: The system installs Maven successfully.

Verify Maven installation:

Output:

Apache Maven 3.8.6 (e.g., version number may vary)

Maven home: /usr/share/maven

Java version: 11.0.9.1, vendor: AdoptOpenJDK, runtime: /usr/lib/jvm/adoptopenjdk-11-hotspot amd64

Default locale: en_US, platform encoding: UTF-8

OS name: "linux", version: "5.4.0-90-generic", arch: "amd64", family: "unix"

Install Gradle:

Output: Gradle archive is downloaded and extracted.

Verify Gradle installation:

Output:

Gradle 7.5.1

Build time: 2022-02-21 11:55:39 UTC

Revision: a1b8b0bcb28fe42893b9bbd8f1b1a2724a98acb0

Kotlin: 1.5.30

Groovy: 3.0.9

Ant: Apache Ant(TM) version 1.10.11 compiled on September 27 2021

JVM: 11.0.9.1 (AdoptOpenJDK 11.0.9.1+1)

OS: Linux 5.4.0-90-generic amd64

Experiment No: 2. Working with Maven: Creating a Maven Project, Understanding the POM File, Dependency Management and Plugins.